

## **PHASE 2 – REQUIREMENT ANALYSIS**

### **1. INTRODUCTION**

The Requirement Analysis phase identifies the functional and technical requirements of the project “**Script-Controlled ACL – Restrict Record Access Based on Field Value.**”

The main purpose is to define how ServiceNow should restrict access to **Institution Details** records based on user roles and the **Branch** field.

### **2. FUNCTIONAL REQUIREMENTS**

#### **2.1 User Requirement**

An EEE test user shall be created with the following details:

| <b>Field</b> | <b>Value</b>      |
|--------------|-------------------|
| User ID      | EEE User          |
| First Name   | EEE               |
| Last Name    | User              |
| Email        | eeeuser@gmail.com |

#### **2.2 Role Requirements**

Four custom roles shall be created:

| <b>Role</b> | <b>Purpose</b> |
|-------------|----------------|
| bb1         | Read           |
| bb2         | Create         |
| bb3         | Write          |
| bb4         | Delete         |

These roles will be used to control different operations on the Institution Details table.

### 3. TABLE REQUIREMENTS

A custom table named **Institution Details** shall be created.

**Table Name:** u\_institution\_details

The table shall contain:

| Field               | Type              |
|---------------------|-------------------|
| Student Roll Number | Auto Number       |
| Student Name        | Reference – User  |
| Faculty Name        | Reference – User  |
| Branch              | Choice            |
| Email               | String            |
| Phone Number        | String            |
| Description         | Multi-line String |

The **Branch** field shall contain:

- ECE
- EEE
- CSE

Multiple records shall be created using different branch values for testing.

### 4. ACL REQUIREMENTS

Four ACLs shall be configured for the Institution Details table.

| ACL        | Operation | Required Role |
|------------|-----------|---------------|
| Read ACL   | Read      | bb1           |
| Create ACL | Create    | bb2           |
| Write ACL  | Write     | bb3           |
| Delete ACL | Delete    | bb4           |

The Read ACL shall also use the condition **Branch is EEE** and an advanced script.

The script shall allow administrators full access and evaluate the required role for normal users.

## 5. ACCESS REQUIREMENTS

The intended access model is:

| User Roles            | Read | Create | Write | Delete |
|-----------------------|------|--------|-------|--------|
| bb1                   | ✓    | —      | —     | —      |
| bb1 + bb2             | ✓    | ✓      | —     | —      |
| bb1 + bb2 + bb3       | ✓    | ✓      | ✓     | —      |
| bb1 + bb2 + bb3 + bb4 | ✓    | ✓      | ✓     | ✓      |
| Administrator         | ✓    | ✓      | ✓     | ✓      |

The actual effective access will depend on the ACL configuration and ServiceNow ACL evaluation.

## 6. SECURITY REQUIREMENTS

The system shall:

- Restrict unauthorized users from accessing protected records.
- Use roles to control individual operations.
- Use the Branch field for record-level access control.
- Allow administrators full access.
- Prevent unauthorized creation, modification, and deletion.

## 7. SOFTWARE REQUIREMENTS

- ServiceNow development/lab instance
- Administrator access
- security\_admin role for ACL configuration
- Web browser
- Internet connection

## 8. TESTING REQUIREMENTS

The ACL configuration shall be tested using different users.

The following cases must be verified:

1. User with bb1 can view permitted EEE records.
2. User without the required role cannot access protected records.
3. User with bb1 + bb2 can create records.
4. User with bb1 + bb2 + bb3 can edit records.

5. User with all four roles can delete records.
6. Administrator can access records without the same restrictions.

## **9. EXPECTED OUTCOME**

After completing the requirements, the system should provide:

- A configured Institution Details table.
- Required custom roles.
- Read, Create, Write, and Delete ACLs.
- Branch-based record restriction.
- Controlled access for different users.
- Full administrative access.

## **10. CONCLUSION**

The Requirement Analysis phase defines the necessary users, roles, table structure, fields, ACL operations, and security requirements for the project.

These requirements provide the foundation for designing the ACL structure and access flow in the next phase.